MATH 345 Skeleton Notes
Unit 6: Constraints and SVD
Section 6.3: Maximum stretch and singular values

One way to ask what a matrix map does is to ask which unit input is stretched the most:

$$\max_{\|\mathbf{x}\|=1} \|A\mathbf{x}\|^2.$$

The square is included because

$$\begin{aligned}\|A\mathbf{x}\|^2 & \\ &= (A\mathbf{x}) \cdot (A\mathbf{x}) \\ &= \mathbf{x}^T A^T A\mathbf{x}.\end{aligned}$$

Thus a question about a matrix map becomes a constrained optimization problem for the scalar-valued function

$$\mathbf{x} \mapsto \mathbf{x}^T A^T A\mathbf{x}$$

on the unit sphere.

Fact Let B be a symmetric matrix, and define

$$q(\mathbf{x}) = \mathbf{x}^T B\mathbf{x}.$$

The extrema of $q(\mathbf{x})$ subject to $\|\mathbf{x}\| = 1$ occur at eigenvectors of B. The extreme values are eigenvalues of B.

Proof Use the constraint

$$g(\mathbf{x}) = \mathbf{x} \cdot \mathbf{x} = 1.$$

For symmetric B,

$$\nabla q(\mathbf{x}) = 2B\mathbf{x}, \qquad \nabla g(\mathbf{x}) = 2\mathbf{x}.$$

The Lagrange multiplier condition gives

$$2B\mathbf{x} = \lambda 2\mathbf{x},$$

so

$$B\mathbf{x} = \lambda \mathbf{x}.$$

Thus constrained critical points occur at eigenvectors of B. If $\mathbf{x}$ is a unit eigenvector with $B\mathbf{x} = \lambda\mathbf{x}$, then

$$\begin{aligned} q(\mathbf{x}) &= \mathbf{x}^T B\mathbf{x} \\ &= \mathbf{x}^T(\lambda\mathbf{x}) \\ &= \lambda\|\mathbf{x}\|^2 \\ &= \lambda. \end{aligned}$$

Since the unit sphere is closed and bounded, the extreme value theorem guarantees that the extrema occur among these candidates.

Apply the fact to $B = A^T A$. The eigenvalues of $A^T A$ are nonnegative, so we can take their square roots. The singular values of A are

$$\sigma_i = \sqrt{\lambda_i},$$

where λ_i are the eigenvalues of $A^T A$, listed so that

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq 0.$$

A unit eigenvector of $A^T A$ is a right singular vector of A. The number σ_i is the stretch factor in that input direction.

Activity: A Small Stretch Computation

Let

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

Let

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

1. Compute $A\mathbf{e}_1$ and $A\mathbf{e}_2$.
2. Compute $\|A\mathbf{e}_1\|$ and $\|A\mathbf{e}_2\|$.
3. Compute $A^T A$.
4. Find the eigenvalues of $A^T A$.
5. What are the singular values of A?
6. Which unit direction is stretched most?

Activity: Reading the same stretch in code

The following code repeats the previous activity using $A^T A$.

```
import numpy as np

A = np.array([[3., 0.],
              [0., 1.]])
```

```
B = A.T @ A
eigvals, eigvecs = np.linalg.eigh(B)

singular_values = np.sqrt(eigvals[::-1])
dominant_direction = eigvecs[:, -1]

B, eigvals, singular_values, dominant_direction
```

Output:

```
(array([[9., 0.],
        [0., 1.]]),
 array([1., 9.]),
 array([3., 1.]),
 array([1., 0.]))
```

1. What mathematical matrix is stored in `B`?
2. Why are the entries of `eigvals` squared stretch factors?
3. Why does the code take a square root?
4. Why does `eigvecs[:, -1]` give the most-stretched input direction?
5. What value of $\max_{\|\mathbf{x}\|=1} \|A\mathbf{x}\|$ does the code predict?

The next section packages these ingredients into one factorization. The right singular vectors are special input directions. The singular values are stretch factors. Multiplying a right singular vector by A gives the corresponding output direction.